Unit 1: Introduction to Product Design and Development

2-Mark Questions:

1. What is the purpose of product design in the development process?
2. Name three modern approaches to product design.
3. Define specialization in the context of product design.
4. What is meant by product validation?

4-Mark Questions:

1. Explain how standardization benefits product design with a relevant example.
2. Discuss the key differences between simplification and specialization in product design.
3. Describe the role of iterative processes in modern product development.

6-Mark Questions :

1. Apply the concept of simplification to design a user-friendly coffee maker. Provide an example.
2. Demonstrate how product testing can be used to ensure the quality of a smartwatch.

12-Mark Questions:

1. Analyze the impact of combining agile and lean methodologies in the product development of a smart home speaker, highlighting potential benefits and drawbacks.
2. Create a detailed product development roadmap for a foldable bicycle, emphasizing testing and validation strategies.

Unit 2: Customer-Centric Product Development

2-Mark Questions:

1. What is the role of a mission statement in guiding product development?
2. List two methods for identifying customer needs.
3. Define technology forecasting in product development.
4. What is customer population analysis?

4-Mark Questions:

1. Explain how the S-curve is used to predict technology trends in product development.
2. Discuss the importance of customer satisfaction in shaping product design decisions.
3. Describe how market segmentation influences product development strategies.

6-Mark Questions:

1. Apply customer needs assessment techniques to design a travel app for budget travelers.
2. Demonstrate how technology forecasting can guide the development of a wearable health device.

12-Mark Questions :

1. Evaluate the challenges of balancing customer needs with technical feasibility in designing an affordable electric car, proposing solutions to address them.
2. Create a customer involvement strategy for developing a smart kitchen appliance, incorporating tools like interviews and user testing.

Unit 3: Concept Development and Design Methodology

2-Mark Questions :

1. What is concept generation in product design?
2. Define the purpose of concept scoring.
3. Name two tools used in design evaluation.
4. What is the subtract-operate procedure?

4-Mark Questions :

1. Explain the importance of lateral thinking in concept development.
2. Discuss how Pugh's concept selection charts facilitate design decisions.
3. Describe the process of concept embodiment in product design.

6-Mark Questions:

1. Apply brainstorming techniques to generate ideas for a sustainable water purifier.

2. Demonstrate the use of simulation-driven design in developing a drone for delivery services.

12-Mark Questions:

1. Analyze the effectiveness of concept generation methods like brainstorming versus structured approaches like TRIZ for designing a modular smartphone, recommending the better approach.
2. Create a concept selection framework for a wearable fitness band, integrating Pugh's charts and technical feasibility analysis.

Unit 4: Product Analysis and Strategic Development

2-Mark Questions :

1. What is the purpose of product teardown analysis?
2. Define trend analysis in product development.
3. Name two applications of reverse engineering.
4. What is a product portfolio?

4-Mark Questions :

1. Explain the benchmarking process in product development with an example.
2. Discuss the role of function-form diagrams in strategic product design.
3. Describe how product specifications are set using trend analysis.

6-Mark Questions :

1. Apply the product teardown process to improve the design of a gaming console.
2. Demonstrate how benchmarking can enhance the features of a smart thermostat.

12-Mark Questions:

1. Evaluate the impact of reverse engineering and benchmarking on the development of a competitive electric bicycle, analyzing their strengths and limitations.
2. Create a product portfolio strategy for a startup launching a range of smart home devices, incorporating trend analysis and function-form diagrams.

Unit 5: Sustainable Design and Product Life Cycle Management

2-Mark Question:

1. What is design for sustainability?
2. Define product data management.
3. Name two aspects of design failure mode effect analysis.
4. What is the weighted sum assessment method?

4-Mark Questions:

1. Explain the concept of design for robustness with an example.
2. Discuss the importance of life cycle assessment in sustainable design.
3. Describe the role of global design considerations in product development.

6-Mark Questions :

1. Apply design for safety principles to develop a child-friendly toy.
2. Demonstrate how product life cycle management can be used to optimize a reusable water bottle.

12-Mark Questions:

1. Analyze the challenges of implementing design for environment principles in the production of budget laptops, proposing strategies to ensure sustainability and affordability.
2. Create a product life cycle management plan for a solar-powered lamp, integrating design failure mode effect analysis and regional design considerations.